

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence
CSA17

COURSE CODE:

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Hour

Time Duration: 1

QUESTIONS: 5
Marks:30

Total

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I __S.Jotheeswaran_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Jotheeswaran

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.

2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. **If a person is a library member, then the person can borrow books.**
2. **Priya is a library member.**

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. **Represent the knowledge base using propositional logic.**
2. **Convert each statement into CNF.**

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. **If a student clears the aptitude test, then the student is eligible for placement.**
2. **Arun cleared the aptitude test.**

Goal

1. **Prove that Arun is eligible for placement using the Resolution Algorithm.**

Tasks

1. **Convert the statements into propositional logic.**
2. **Write the CNF clauses.**
3. **Negate the goal.**
4. **Apply the Resolution Algorithm step by step.**
5. **Draw the resolution tree and conclude.**

Question 5 – Access Control System

Knowledge Base

- 1. If a user enters the correct password, then the user is authenticated.**
- 2. If a user is authenticated, then the user is granted access.**
- 3. The user entered the correct password.**

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

- 1. Represent the knowledge base in propositional logic.**
- 2. Convert all implications into CNF.**
- 3. Negate the goal.**
- 4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).**
- 5. Explain each resolution step and write the final conclusion.**

1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1: Rain and Wet Ground

Theoretical Foundation of Resolution

Resolution is a sound and complete inference mechanism in propositional logic. It relies on the principle of proof by contradiction (refutation). To prove a goal, the knowledge base (KB) is converted into Conjunctive Normal Form (CNF), the goal is negated and added to the KB, and the resolution rule is repeatedly applied to derive an empty clause, which signifies a logical contradiction.

Task 1: Representation in Propositional Logic

Let us define the propositional variables for the given statements:

- R: It rains
- W: The ground becomes wet

Knowledge Base (KB):

1. $R \rightarrow W$ (If it rains, then the ground becomes wet)
2. R (It is raining)

Task 2: Conversion to Conjunctive Normal Form (CNF)

To convert the implication into CNF, we apply the material implication rule ($A \rightarrow B \equiv \neg A \vee B$).

Clause 1 (C1): $\neg R \vee W$

Clause 2 (C2): R

Task 3 & 4: Negating the Goal and Applying Resolution Algorithm

Goal: W (The ground is wet)

Negated Goal (C3): $\neg W$

We now add the negated goal to our clause set {C1, C2, C3} and apply the resolution rule:

Step 1: Resolve C1 ($\neg R \vee W$) with C2 (R).

- The complementary literals $\neg R$ and R cancel each other out.
- Resolvent 1 (R1): W

Step 2: Resolve R1 (W) with the negated goal C3 ($\neg W$).

- The complementary literals W and $\neg W$ cancel each other out.
- Resolvent 2: Empty Clause ($\square$)

Conclusion

Since the application of the resolution algorithm resulted in the empty clause ($\square$), a contradiction has been reached. Therefore, the negated goal is false, proving that the original goal 'W' (the ground is wet) logically follows from the Knowledge Base.

Question 2: Student Assignment Submission

Task 1: Representation in Propositional Logic

We define the specific propositional variables relevant to the student (Rahul):

- S: Rahul submits the assignment
- M: Rahul receives internal marks

Knowledge Base (KB):

1. $S \rightarrow M$ (If a student submits the assignment, then the student receives internal marks)
2. S (Rahul submitted the assignment)

Task 2: Conversion to CNF

Eliminate the implication for the first premise:

Clause 1 (C1): $\neg S \vee M$

Clause 2 (C2): S

Task 3 & 4: Negating the Goal and Applying Resolution

Goal: M (Rahul receives internal marks)

Negated Goal (C3): $\neg M$

Resolution Steps:

1. Take C1 ($\neg S \vee M$) and C2 (S). Resolving these by canceling the 'S' literal yields a new clause, R1: M.
2. Take the derived clause R1 (M) and resolve it with the negated goal C3 ($\neg M$).
3. The literals M and $\neg M$ are exact complements, leading to their mutual elimination. The result is the empty clause ($\square$).

Conclusion

The derivation of the empty clause confirms that assuming Rahul does not receive internal marks contradicts the established facts. Thus, it is proved that Rahul receives internal marks.

Question 3: Library Membership

Task 1: Representation in Propositional Logic

Let us define the propositions for Priya's case:

- L: Priya is a library member
- B: Priya can borrow books

Knowledge Base (KB):

1. $L \rightarrow B$

2. L

Task 2: Conversion to CNF

Using material implication:

Clause 1 (C1): $\neg L \vee B$

Clause 2 (C2): L

Task 3: Negating the Goal and Resolution

Goal: B (Priya can borrow books)

Negated Goal (C3): $\neg B$

Resolution Iteration:

- Resolving C1 ($\neg L \vee B$) and C2 (L): This eliminates L and leaves the resolvent B.
- Resolving the new resolvent (B) with C3 ($\neg B$): This eliminates B completely, resulting in no remaining literals.

Conclusion

No new clauses can be generated, and the final resolvent is the empty clause ($\square$). This proves absolutely that Priya can borrow books based on the given knowledge base.

Question 4: Placement Eligibility

Task 1 & 2: Representation and CNF Conversion

Variables for Arun:

- A: Arun clears the aptitude test
- E: Arun is eligible for placement

Knowledge Base & CNF Clauses:

Premise 1: $A \rightarrow E \implies C1: \neg A \vee E$

Premise 2: $A \implies C2: A$

Task 3 & 4: Negated Goal and Resolution Steps

Goal: E (Arun is eligible for placement)

Negated Goal (C3): $\neg E$

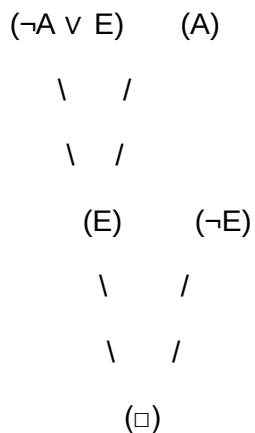
Resolution Refutation:

Step 1: Resolve C1 ($\neg A \vee E$) with C2 (A) to derive E.

Step 2: Resolve the derived E with C3 ($\neg E$) to derive $\square$ (Empty Clause).

Task 5: Resolution Tree and Conclusion

Resolution Tree Visualization:



Conclusion: The resolution tree successfully terminates in an empty clause, providing a rigorous logical proof that Arun is eligible for placement.

Question 5: Access Control System

Task 1: Representation in Propositional Logic

This system requires a transitive logical chain. We define:

- P: The user enters the correct password
- A: The user is authenticated
- G: The user is granted access

Knowledge Base:

1. $P \rightarrow A$
2. $A \rightarrow G$
3. P

Task 2: Conversion of All Implications to CNF

Applying the material implication rule ($X \rightarrow Y \equiv \neg X \vee Y$) to both rules:

Clause 1 (C1): $\neg P \vee A$

Clause 2 (C2): $\neg A \vee G$

Clause 3 (C3): P

Task 3: Negating the Goal

Goal: G (The user is granted access)

Negated Goal (C4): $\neg G$

Task 4 & 5: Applying Resolution and Final Conclusion

We add C4 to our KB and systematically resolve clauses:

Step 1: Resolve C1 ($\neg P \vee A$) with C3 (P).

- The contradictory literal is P. Resolving them yields the resolvent R1: A.
- Logical interpretation: Since the user entered the password, they are authenticated.

Step 2: Resolve C2 ($\neg A \vee G$) with the newly derived R1 (A).

- The contradictory literal is A. Resolving them yields the resolvent R2: G.
- Logical interpretation: Since the user is authenticated, they are granted access.

Step 3: Resolve R2 (G) with the negated goal C4 ($\neg G$).

- The complementary literals G and $\neg G$ cancel perfectly.

- Result: Empty Clause ($\square$).

Conclusion: The step-by-step resolution uncovers a contradiction when we assume the user is denied access ($\neg G$). Therefore, by resolution refutation, we logically conclude that the user is granted access.